

YU ZHAO

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EDUCATION

Pompeu Fabra University (UPF) & Barcelona Institute for Global Health (ISGlobal)

Ph.D. in Public Health (Environmental Epidemiology)

Sep 2023 – Present

Barcelona, Spain

- **Research focus:** Prenatal heat exposure, placental function and epigenetics, fetal brain development, and early childhood neurodevelopment.
- **Supervisors:** Payam Dadvand, Mariona Bustamante, and Marta Cosin.

Sun Yat-sen University

Master of Medicine in Maternal, Child and Adolescent Health

Sep 2020 – Jul 2023

Guangzhou, China

- **Master's thesis:** Greenspace exposure and visual impairment among Chinese schoolchildren.
- **Supervisor:** Yajun Chen.

Anhui Medical University

Bachelor of Medicine in Preventive Medicine

Sep 2015 – Jul 2020

Hefei, China

RESEARCH PROFILE

Environmental epidemiology PhD researcher specializing in environmental exposure assessment and maternal and child health. Experienced in applying machine-learning and GIS-based methods to estimate individual-level environmental exposures and analyzing prospective cohort data in R. Current research examines the effects of prenatal heat exposures on placental DNA methylation, fetal brain morphology, and early childhood neurodevelopment.

PUBLICATIONS

- **A two-stage modeling approach to estimate indoor NO₂ exposure: a Barcelona case study** 
Alan Domínguez, **Yu Zhao**, Karl Samuelsson, Pol Jimenez-Arenas, *et al.*, Ioar Rivas, and Xavier Basagaña.
Journal of Exposure Science & Environmental Epidemiology. 2026. Advance online publication.
- **Prediction of daily home indoor temperature and relative humidity using a deep ensemble machine learning approach** 
Yu Zhao, Alan Domínguez, Karl Samuelsson, Toni Galmes, *et al.*, Ioar Rivas, and Payam Dadvand.
Building and Environment. 2026;295:114392.
- **Air pollution and fetal brain morphological development: a prospective cohort study** 
Laura Gómez-Herrera*, **Yu Zhao***, Ioar Rivas, Elisenda Eixarch, *et al.*, Payam Dadvand, and Jordi Sunyer.
The Lancet Planetary Health. 2025;9(6):e480–e490.
- **Urban greenspace and visual acuity in schoolchildren: a large prospective cohort study in China** 
Wen-Wen Bao*, **Yu Zhao***, Payam Dadvand, Nan Jiang, *et al.*, Guanghui Dong, and Yajun Chen.
Environment International. 2024;184:108423.
- **Association between greenspace and blood pressure: a systematic review and meta-analysis** 
Yu Zhao*, Wen-Wen Bao*, Bo-Yi Yang, Jing-Hong Liang, *et al.*, Guang-Hui Dong, and Ya-Jun Chen.
Science of the Total Environment. 2022;817:152513.
- **Face-to-face physical activity incorporated into dietary intervention for overweight/obesity in children and adolescents: a Bayesian network meta-analysis** 
Jing Hong Liang, **Yu Zhao**, Yi Can Chen, *et al.*, Aertzguli Kakaer, and Ya Jun Chen.
BMC Medicine. 2022;20:325.
- **Development and validation of a nomogram-based prognostic model to predict high blood pressure in children and adolescents—findings from 342,736 individuals in China** 
Jing-Hong Liang, **Yu Zhao**, Yi-Can Chen, *et al.*, Aertzguli Kakaer, and Ya-Jun Chen.
Frontiers in Cardiovascular Medicine. 2022;9:884508.

* Co-first authors who contributed equally.

RESEARCH EXPERIENCE

Barcelona Institute for Global Health (ISGlobal) & Pompeu Fabra University

Ph.D. Researcher in Environmental Epidemiology

Sep 2023 – Present

Barcelona, Spain

- **Deep ensemble machine-learning exposure modelling.** Developed and validated models to predict daily home indoor temperature and relative humidity throughout pregnancy by integrating repeated indoor measurements with outdoor meteorology, dwelling and neighborhood characteristics, and occupant behaviors.
- **Prenatal heat exposure, fetal brain development, and childhood neurodevelopment.** Investigated associations of prenatal indoor and outdoor heat exposure with fetal brain morphology assessed by third-trimester neurosonography and early childhood neurodevelopment in the Barcelona Life Study Cohort (BiSC). Applied exploratory mediation analysis to assess whether fetal brain morphology may link prenatal heat exposure to later neurodevelopment.
- **Prenatal heat exposure and the placental epigenome.** Analyzed placental DNA methylation data to identify epigenetic signatures associated with prenatal heat exposure and performed functional and disease-enrichment analyses to investigate potential biological pathways.
- **Prenatal heat exposure and autism-related traits.** Examined associations between prenatal heat exposure and autism-related traits assessed using the Modified Checklist for Autism in Toddlers (M-CHAT) at 18 months and the Social Responsiveness Scale, Second Edition (SRS-2), at 48 months, building on autism-related enrichment signals identified in the placental epigenomic analyses.
- **Core methodological expertise.** Apply machine-learning exposure prediction, GIS-based exposure assessment, longitudinal epidemiological analysis, mediation analysis, placental epigenomic analysis, functional enrichment, and reproducible data workflows in R.

Sun Yat-sen University

Graduate Researcher in Maternal, Child and Adolescent Health

Sep 2020 – Jul 2023

Guangzhou, China

- **Greenspace exposure assessment.** Characterized residential greenspace using satellite-derived vegetation indices, including the Normalized Difference Vegetation Index (NDVI), Enhanced Vegetation Index (EVI), and Soil-Adjusted Vegetation Index (SAVI), together with street-view-based visible greenness.

- **Greenspace and childhood visual development.** Investigated associations between multidimensional greenspace exposure and changes in visual acuity and visual impairment over four years of follow-up among Chinese schoolchildren.
- **Potential mediating pathways.** Assessed the potential mediating roles of physical activity and air pollution in the association between greenspace exposure and childhood visual development.
- **Methods.** Applied GIS-based environmental exposure assessment, longitudinal cohort analysis, mediation analysis, epidemiological data management, and statistical analysis in R.

SELECTED CONFERENCE PRESENTATIONS

Oral Presentations

- [Prenatal indoor and outdoor heat exposure, fetal brain morphology, and early childhood neurodevelopment](#)
ISEE Annual Conference, Munich, Germany, Aug 30–Sep 2, 2026.
- [The role of fetoplacental hemodynamics in the association between prenatal traffic-related air pollution and birth weight](#)
ISEE Young 2024, Rennes, France, Jun 5–7, 2024.

Poster Presentations

- [Prediction of daily home indoor temperature and relative humidity using a deep ensemble machine learning approach](#)
ISEE Annual Conference, Munich, Germany, Aug 30–Sep 2, 2026.
- [Prenatal indoor and outdoor heat exposure, fetal brain morphology, and early childhood neurodevelopment](#)
Global Exposome Summit, Sitges, Spain, Apr 27–29, 2026.

HONORS AND AWARDS

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| ◦ Travel Award , ISEE Annual Conference, Munich, Germany | 2026 |
| ◦ First Prize for a Conference Abstract , ISEE Young, Rennes, France | 2024 |
| ◦ First-class Scholarship for Postgraduate Students , Sun Yat-sen University | 2022 |
| ◦ Second-class Scholarship for Postgraduate Students , Sun Yat-sen University | 2021 |
| ◦ Guoyuan Securities Scholarships and Grants | 2018–2020 |
| ◦ National Encouragement Scholarship | 2017 |
| ◦ Zhang Xiqi First-class Scholarship | 2016 |

ACADEMIC SERVICE

- **Oral Session Co-chair**, ISEE Annual Conference, Munich, Germany, 2026.
- **Journal peer review:** *Environment International*; *Environmental Research*; *Environmental Science & Technology*; *Journal of Affective Disorders*; *One Health*; and *Journal of Exposure Science & Environmental Epidemiology*.